GAME3023 Week 3 Notes

Unreal Engine: Development

Week 3 Notes • Arena Blockout — Part One

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UE

combat arena

Week 3

Arena Blockout — Part One

- Begin the two-week arena blockout phase.
- Translate Checkpoint 1 planning into a rough playable environment.
- Use Unreal level tools to establish scale, routes, traversal, verticality, and major spaces.
- Test early with the player character, not only from the editor camera.
- Prepare for Milestone 1: First Playable, due Week 4.

Semester Roadmap

The same project grows from concept to release.



Week 3 focus

Turn the plan into the first rough in-engine version of the arena.

Milestone 1 is due next week. The arena does not need combat yet, but it must be playable enough to evaluate.

Part 1

Where we are this week

Checkpoint 1 should now become an Unreal level, not just a plan.

This Week's Purpose

- Not to finish the arena.
- Not to solve combat, networking, AI, UI, or scoring yet.
- Not to make the arena beautiful.
- To build the rough playable space so scale, movement, routes, and flow can be tested.
- To discover problems while they are still easy to fix.

What Should Already Exist

- Checkpoint 1 has been submitted or is being finalized.
- The Unreal project opens correctly.
- The team has a concept, rough arena plan, and initial feature list.
- The team has chosen or narrowed the opponent path: AI or human/networked.
- The team has a folder/workflow plan and a realistic project scope.

Week 3 Development Mindset

- Build fast enough to learn from the level.
- A blockout is a design tool, not a final art pass.
- Every major space should be testable in Play-In-Editor.
- Expect to revise the layout repeatedly.
- If the arena cannot be tested, it cannot be improved.

From Plan to Playable Space

- Use the Checkpoint 1 sketch as a guide, not a contract.
- Block out the full footprint before polishing individual rooms.
- Prioritize floors, walls, ramps, platforms, openings, and major routes.
- Add placeholders for future hazards, pickups, spawns, and interactive mechanics.
- Make notes when the plan fails in 3D.

Week 3 Output Snapshot

Area	What should be visible by the end of the week
Arena footprint	Rough shape of the entire arena exists in-engine
Major spaces	Main combat areas, side areas, and routes are represented
Player scale	Early PIE testing has confirmed or challenged the scale
Spawn planning	Initial spawn areas are placed or clearly marked
Future systems	Hazard, pickup, interaction, and opponent locations are roughed in

Part 2

Blockout tools and workflow

Use the fastest reliable Unreal tools to build a testable space.

What Is a Blockout?

- A blockout is a rough playable version of a level built with simple shapes.
- It tests size, flow, visibility, player movement, and layout before detail is added.
- It prevents teams from wasting time decorating a broken space.
- For this project, the blockout is the foundation of the entire combat arena.
- The blockout will continue evolving after Milestone 1.

Why Blockout Matters

- Combat only works if the space supports it.
- AI or human opponents need routes, sightlines, and readable encounter spaces.
- Interactive mechanics need enough room to matter.
- Pickups and hazards need meaningful placement.
- A weak arena layout cannot be fixed by adding nicer materials later.

Unreal Level Editor Review

- Use the Viewport to place, move, rotate, and scale objects.
- Use the Outliner to keep track of placed Actors.
- Use the Details panel to adjust transform, collision, materials, and actor settings.
- Use the Content Browser to organize maps, Blueprints, materials, and documentation.
- Save and test the level frequently.

Blockout Tool Options

Tool / workflow	Best use	Caution
Geometry Brushes / BSP-style	Fast floors, walls, ramps, stairs, simple volume layout	Good for early design, not final beauty
Modeling Mode	Create/edit simple geometry directly in Unreal	Do not over-model during blockout
Static Mesh primitives	Reusable simple shapes, platforms, blockers, cover	Keep organized and named clearly
Blueprint placeholders	Mark future doors, platforms, hazards, pickups	Do not spend too long building systems yet

Geometry Brushes and Modeling Mode

Geometry Brushes / BSP-Style

Useful for quick layout and older Unreal-style blockouts.
Good for floors, walls, stairs, ramps, and simple volumes.
Fast to create and modify.

Modeling Mode

Create and edit geometry directly in Unreal.
Useful for primitives, CubeGrid-style construction, simple mesh edits, and shape refinement.

A Practical Tool Rule

- Use whatever gets the level playable fastest.
- Do not choose a complex modeling workflow just because it looks more advanced.
- Simple boxes, ramps, stairs, and platforms are enough for the first pass.
- If a tool slows the team down, switch to a simpler approach.
- The First Playable is judged on function, readability, and testability—not final art.

Grid Snapping Is Your Friend

- Use grid snapping to keep walls, floors, ramps, and platforms aligned.
- Consistent units make collision and player scale easier to judge.
- Snapping helps the team make clean changes later.
- Messy placement creates gaps, overlaps, and hard-to-debug collision issues.
- Turn snapping off only when there is a clear reason.

Working With Simple Shapes

- Start with large primitive forms: floors, boundaries, platforms, ramps, blocks, and cover.
- Build the arena as a collection of readable gameplay spaces.
- Avoid tiny details during the first pass.
- Use duplicated modular shapes to keep the layout consistent.
- Keep geometry simple enough that it can be changed quickly after testing.

Keep the Map Editable

- Avoid turning every early shape into a precious final asset.
- Group or name important blockout pieces clearly.
- Keep major routes easy to modify.
- Do not spend an hour perfecting a wall that may be deleted next week.
- Use placeholders for features that will be built later.

Part 3

Scale, traversal, and the player

The player character is the measuring tool for the whole arena.

Player Scale Comes First

- A space can look good from the editor camera and still feel wrong when played.
- Test with the actual character/camera as early as possible.
- Hallways, doors, ramps, stairs, platforms, and gaps must fit the player.
- Combat arenas need room to move, retreat, dodge, and re-enter fights.
- Scale problems are easiest to fix before the arena has many systems attached.

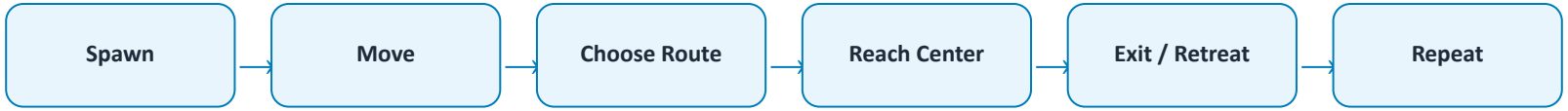
Use the Player Character Early

- Place the player start.
- Press Play-In-Editor.
- Run the intended routes.
- Jump, turn, backtrack, and test corners.
- Adjust geometry immediately when something feels wrong.

Traversal Questions

- Can the player reach every important space?
- Can the player understand where to go next?
- Are ramps, stairs, lifts, and platforms comfortable to use?
- Are there routes for retreat and re-entry after combat?
- Does movement feel too slow because the arena is too large?
- Does the space feel cramped because the arena is too tight?

First-Pass Traversal Loop



A good blockout lets the player naturally circulate through the arena.
If the player only runs down one corridor, the design needs more route thinking.

Scale Testing Pass

- Start from each planned spawn area.
- Time how long it takes to reach central spaces.
- Check whether vertical routes feel useful or awkward.
- Look for dead ends, cramped corners, and oversized empty rooms.
- Record what must change before Milestone 1.

Collision Basics

- The player should not fall through floors or snag constantly on blockout geometry.
- Collision does not need final polish, but the arena must be navigable.
- Test ramps, stairs, platforms, doors, and tight spaces.
- Watch for invisible walls and accidental gaps.
- Bad collision can make a good layout feel broken.

PlayerStarts and Spawn Testing

- Use PlayerStart actors or equivalent spawn markers early.
- Test the player's first view when gameplay begins.
- Avoid spawning directly into walls, pits, hazards, or immediate unavoidable danger.
- In multiplayer, spawn fairness will matter later.
- In AI projects, spawn placement still affects pacing and encounters.

Part 4

Building the arena layout

Construct the big playable structure before individual feature details.

Build Major Spaces First

Layer	Week 3 priority
Shape	Create the overall footprint and major spaces
Scale	Test with the player character early
Routes	Connect spaces with more than one meaningful path
Verticality	Add upper/lower routes where they serve gameplay
Spawns	Mark preliminary player/opponent spawn areas
Hazards / Pickups	Use placeholders for later feature development

Start With the Whole Arena

- Block out the overall footprint first.
- Add the main combat areas before decorating side rooms.
- Connect the major spaces with rough routes.
- Make sure there is a complete navigable loop.
- A complete rough arena is more useful than one polished corner.

Major Space Types

- Spawn areas: where players or opponents enter play.
- Central conflict area: where players are likely to meet.
- Side routes: alternate movement paths around danger or conflict.
- High ground: advantageous but exposed positions.
- Resource areas: locations that may later hold pickups or weapons.
- Hazard areas: places where the environment creates danger.

Connecting Routes

- Routes should create choices.
- Avoid making every path equally safe, equally long, and equally boring.
- Connect upper and lower areas where possible.
- Give players more than one way to enter and exit important spaces.
- Dead ends should have a purpose, such as a pickup or risk/reward moment.

Verticality in the Blockout

- Build ramps, stairs, lifts, platforms, ledges, bridges, or jump-pad destinations as simple forms.
- Test whether upper routes are worth taking.
- Avoid height differences that look interesting but do not affect gameplay.
- Make sure vertical movement does not break flow.
- If verticality exists, it should change decisions during combat.

Sightlines and Cover

- Stand in important locations and look across the map.
- Ask what a player can see immediately after spawning.
- Long sightlines can create powerful ranged positions.
- Short sightlines can create ambushes and close combat.
- Cover, walls, elevation, and route bends help control what players can see.

Chokepoints and Open Areas

- A chokepoint is an area where movement narrows and players are forced together.
- Chokepoints can create exciting conflict, but too many can make the map frustrating.
- Open areas allow movement but may expose players too much.
- Mix open, narrow, high, low, safe, and risky spaces.
- The blockout should already suggest this variety.

Central Conflict Area

- Many arenas benefit from a central area where conflict naturally happens.
- The center should usually connect multiple routes.
- It may contain a powerful pickup, hazard, vertical route, or interactive element later.
- Avoid making the center either completely safe or instantly deadly.
- Test whether players naturally move toward it.

Placeholders Are Useful

- Use simple labeled shapes to mark future pickups, hazards, doors, lifts, jump pads, or weapons.
- A placeholder is not a finished feature.
- It tells the team where future gameplay systems are expected to live.
- Placeholders help Joss understand the design intention during reviews.
- They also help prevent forgetting planned features.

Early Spawn Planning

- Mark possible player/opponent spawn zones, even if final spawn logic is not built.
- Test whether each spawn leads to meaningful choices.
- Avoid placing every spawn beside the same resource.
- Avoid making one spawn obviously safer or stronger.
- Spawn placement will be revised after combat and opponent testing.

Team Workflow During Blockout

- Avoid having multiple people edit the same map file at the same time without coordination.
- Decide who is responsible for current map changes.
- Others can prototype mechanics, materials, reference diagrams, or test notes separately.
- Communicate before replacing or reorganizing level geometry.
- Keep backups before major layout changes.

Part 5

Testing and revising the blockout

Blockout is a cycle: build, play, observe, revise.

Why Testing Is a Design Tool



The goal is not to defend the first layout. The goal is to discover what the space needs to become playable.

What to Test Every Pass

- Can the player move through every major area?
- Does the arena have a complete loop?
- Are routes readable from the player perspective?
- Are spaces too large, too small, or too similar?
- Is there potential for interesting combat?
- Are there places where hazards, pickups, or interactive mechanics would matter?

Blockout Review Questions

- Where does the player naturally go first?
- Where will fights happen?
- Where can the player escape when low on health?
- Where could a powerful pickup be placed?
- Where could a hazard create a meaningful decision?
- What part of the map is currently least interesting?

Common Blockout Problems

- The arena is only a rectangle with walls.
- The map is too large and feels empty.
- The map is too cramped for combat.
- There is only one route between important spaces.
- Verticality exists but has no gameplay purpose.
- Spawn locations are unfair or unclear.
- Future features have no planned space.

Fast Revision Strategies

- Move large shapes before adding details.
- Delete or combine spaces that do not serve gameplay.
- Add a second route before adding decorative detail.
- Use temporary colour/materials to mark gameplay meaning.
- Make one change, test it, then decide whether it helped.

Keep a Blockout Issue List

- Scale problems.
- Collision problems.
- Confusing routes.
- Dead or boring areas.
- Spawn concerns.
- Possible pickup/hazard locations.
- Technical questions for Joss.
- Features to cut if the arena grows too large.

Part 6

Preparing for Milestone 1

Milestone 1 is First Playable, due next week.

Milestone 1: First Playable

- Due Week 4.
- The arena blockout should be complete enough to run through and evaluate.
- Player scale, traversal, routes, collision, and spawn planning should be visible.
- Basic materials/lighting should begin to communicate spaces and gameplay meaning.
- Combat is not required yet; the focus is the playable arena foundation.

What First Playable Should Include

- A complete rough arena layout.
- Major spaces and routes.
- Player can run through the intended playable area.
- Early verticality where appropriate.
- Preliminary spawn locations.
- Collision reliable enough for testing.
- Basic materials, lighting, or colour coding for readability.
- At least one early interactive feature under development or represented clearly.

What First Playable Does Not Need Yet

- Finished combat.
- Finished AI or networking.
- Final UI.
- Final lighting and art polish.
- Complete pickups or scoring.
- Every planned hazard fully functional.
- Perfect map balance.
- Final geometry decisions.

Milestone 1 Grading Focus

- Is the arena actually playable and testable?
- Does the scale work from the player perspective?
- Are routes, spaces, and verticality visible?
- Is collision acceptable for early testing?
- Are future gameplay features planned within the level?
- Is the project organized enough to continue development?

Submission Preparation

- Confirm the project opens on another machine or after a clean restart.
- Save the main map and ensure it is clearly named.
- Remove unnecessary temporary files according to submission instructions.
- Include any required notes, screenshots, or explanation requested by Joss.
- Do not submit a version that only works on one team member's computer.

Part 7

Week 3 work session

Use class time to make the arena real.

Week 3 In-Class Work Plan

- Open the project and review the Checkpoint 1 plan.
- Create or confirm the main arena map.
- Build the rough footprint of the entire arena.
- Add major floors, boundaries, ramps/platforms, and routes.
- Place a PlayerStart and test player scale.
- Add rough placeholders for future spawns, pickups, hazards, and interactions.
- Record layout problems for revision.

Suggested Team Task Split

- Student A: builds and revises the major arena geometry.
- Student B: tests scale/traversal, marks collision issues, and documents route problems.
- Student C: prepares placeholder materials, labels, spawn/pickup/hazard markers, and future feature notes.
- Rotate responsibilities so everyone understands the map.

If Working Individually

- Keep the arena smaller and more focused.
- Build the whole rough layout before adding complexity.
- Use fewer but more meaningful routes and vertical elements.
- Track open problems carefully so you can ask targeted questions.
- Do not attempt a three-person scope alone.

Ask Joss If…

- You cannot decide whether the arena is too large or too small.
- You are unsure which Unreal tool is best for your blockout.
- Player scale feels wrong and you cannot identify why.
- Team members need help coordinating map edits.
- Your planned mechanics do not seem to fit the physical space anymore.

Useful Documentation Links

Use these as starting points during this week.

- Level Editor — editor layout, viewport, Details, Outliner, and Content Browser.
- Modeling Mode — creating and editing level geometry directly in Unreal.
- Geometry Brush Actors — BSP/brush-style blockout workflows and brush concepts.
- Viewport Controls — moving around the 3D viewport and manipulating objects.
- Actors — objects placed in the level and how Unreal treats them.
- Player Start — starting location and orientation for players in the level.

Official UE documentation links should also be posted in Brightspace/class notes for easy access.

AI / Development Assistance This Week

- Ask AI to explain a tool or workflow, not to replace your judgment.
- Useful prompts: “What is a blockout in Unreal level design?” or “How do Geometry Brushes differ from Modeling Mode?”
- Ask for checklists: “What should I test in a first-person arena blockout?”
- Verify all advice in Unreal and through official documentation.
- Record meaningful AI/tool assistance in your development notes.

Common Week 3 Pitfalls

- Spending too much time on small details.
- Building only one room instead of the whole rough arena.
- Floating in the editor camera instead of testing as the player.
- Ignoring collision problems until next week.
- Forgetting to place spawn areas and future gameplay placeholders.
- Treating the Checkpoint 1 sketch as unchangeable.

Part 8

End of Week 3

The arena should now exist as a rough playable space.

Main Takeaway

First Playable starts with the space, not the systems.

- Build the arena quickly enough to test it.
- Test from the player’s perspective.
- Revise scale and routes before adding too many details.
- Leave Week 3 with something that can become Milestone 1 next week.

Before Next Class

- Rough arena blockout exists in Unreal.
- Player can run through major spaces.
- Spawn areas are placed or clearly marked.
- Routes, verticality, and collision have been tested.
- Basic materials or colour coding are started.
- Placeholder locations for hazards, pickups, interactions, and opponents are visible.
- Open issues are recorded.

Week 3 End

- Next week: Arena Blockout — Part Two.
- Focus: refine the blockout into Milestone 1: First Playable.
- Expect to revise scale, routes, collision, visual readability, and early interaction.
- Come ready to show the arena in Play-In-Editor.